Hitelecom®

HIGH-FREQUENCY VIBRATION SENSOR

H Series | Cabled External Probe

H200 LoRa | H300 4G | H500 NB



KEY INDICATORS

Measuring Range

0-50 mm/s RMS
10-5,000 Hz
1-axis / 3-axis

Accuracy

1% reference*
80 Hz calibration

Wireless

4G / NB-IoT
/ LoRa

Protection

IP68
Transmitter enclosure

Battery Life

Over 10 years*
1-hour reporting
Configured duty

OVERVIEW

The Hitelecom H Series high-frequency vibration sensor combines a cabled external probe with a wireless terminal for machinery monitoring. Mount the probe firmly on a rigid motor or bearing-housing point and send configured readings to Hitelecom Platform or an agreed customer platform. Select single-axis or triaxial sensing, with a velocity RMS range of 0-50 mm/s, a display resolution of 0.1 mm/s and a measurement frequency range of 10-5,000 Hz. Match H200 LoRa, H300 4G or H500 NB-IoT and the acquisition / reporting profile to the installation.

FEATURES

CLOUD

Send vibration records to Hitelecom Platform or an agreed customer / private platform.

GREEN

Over 10 years at 1-hour reporting.*
Life depends on configuration.

IP68

IP68 wireless transmitter enclosure. Match probe, cable and connector protection to the installation.

PROBE AT MACHINE

Mount the external probe on a rigid motor bearing housing. Secure the terminal separately.

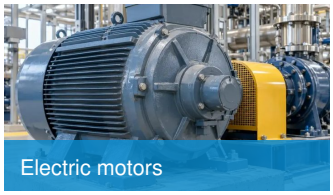
NFC / OTA

NFC local configuration and remote OTA updates on supported configurations.

AXES & RMS

Single-axis or triaxial sensing. Velocity RMS display resolution: 0.1 mm/s.

APPLICATIONS


Electric motors


Pump stations


Fans & ventilation


Air compressors


Gearboxes & conveyors


Machine tools


Heavy processing

Representative machinery environments; mount the probe at a defined rigid motor or bearing-housing point.

Item		Description
Physical Characteristics	Product models	H200 (LoRa) / H300 (4G) / H500 (NB-IoT)
	Power supply	Battery-powered; battery / power option selected by configuration.
	Battery life	Over 10 years at 1-hour reporting; configuration / duty dependent.*
	SIM card	H300 / H500: IoT SIM or eSIM option; H200: not required.
	Antenna	Built In
	Size	Terminal, probe and cable dimensions by configuration.
	Terminal mounting	Mount separately for practical access and radio coverage.
Connectivity	Wireless options	H200: LoRa; H300: 4G; H500: NB-IoT.
	Regional radio settings	LTE / NB-IoT bands and LoRa channel plan by selected variant.
	Data protocol	MQTT platform interface; H200 connects via a compatible gateway.
Vibration Measurement	Velocity RMS range	0-50 mm/s
	Frequency range	10-5,000 Hz (measurement frequency band)
	Sensing axes	Single-axis or triaxial, by configuration.
	Velocity RMS display resolution	0.1 mm/s (display increment)
	Accuracy reference	1% at 80 Hz calibration; supplied probe specification.*
	Operating context	Record machine speed, load and measurement direction.
Probe Assembly	Probe mounting	Rigid motor / bearing housing at a defined stationary point.
	Cable routing	Provide strain relief; route clear of rotating parts and power leads.
	Acquisition duty	Measurement interval / duration and reporting by configuration.

SPECIFICATIONS

Item		Description
Waterproof Rating	Physical body protection	IP68 (wireless transmitter enclosure)
	Ingress protection	IP67/IP68
Reliability	Electrostatic discharge immunity	Contact discharge: 6 kV; air discharge: 8 kV
	Surge immunity	Differential mode: 1 kV; common mode: 2 kV
	Fast transient burst	Power supply circuit: 2 kV
Environmental Range	Operating temperature	-20°C to +70°C (published vibration reference)
	Operating humidity	5% to 95% RH (non-condensing; platform reference)
	Storage temperature	-20°C to +80°C (published vibration reference)
Integration	Data destination	Hitelecom Platform or an agreed customer / private platform.
	LoRa integration	H200 uses a compatible LoRa receiver / gateway.
	Threshold alerts	Define velocity RMS limits and the inspection / notification workflow.
Certification References CCC / CE / FCC / RoHS	Safety	EN IEC 62368-1:2020+A11:2020
	Health	EN IEC 62311:2020
	EMC	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-52 V1.2.1 (2021-11)
	Radio	ETSI EN 301 908-1 V15.2.1 (2023-01) ETSI EN 301 908-13 V13.2.1 (2022-02)
	Frequency range	FDD B1: 1920-1980 MHz TX; 2110-2170 MHz RX FDD B3: 1710-1785 MHz TX; 1805-1880 MHz RX FDD B8: 880-915 MHz TX; 925-960 MHz RX TDD B40: 2300-2400 MHz TX / RX
	Maximum output power	FDD B1: 22.89 dBm max. FDD B3: 23.23 dBm max. FDD B8: 24.02 dBm max. TDD B40: 22.24 dBm max.
	Modulation	QPSK, 16QAM

* Accuracy: 1% at 80 Hz calibration. RMS display resolution is separate; full-band probe accuracy follows configuration.

* Battery life: published reference over 10 years at 1-hour reporting. Actual life depends on battery, probe acquisition duty, reporting schedule, radio coverage and temperature.

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